

INDIAN STATISTICAL INSTITUTE

Project Linked Person — Selection 2026–2027

Practical Examination

Total: $14 + 16 + 10 = 40$ marks Duration: 90 minutes

EXAMINATION INSTRUCTIONS

1. You have root on a virtual machine. It is connected to no real network. Log in with `ssh candidate@192.168.56.10`.
2. Each question has four parts: *the problem*, *how to reproduce it*, *what to fix*, and *how to check your work* (so that you do not have to guess whether you are finished).
3. Partial credit would be given if you can explain the problem and the reason behind, even if you cannot fix it.

Q1. Disk full on the lab file server.

[14 marks]

Reported by Debarshi, research scholar: “I cannot copy anything into `/srv/data`, it says the disk is full, and nothing I can see in there accounts for that much space. Submitting a job into `/srv/spool` *also* says the disk is full, but `df -h` shows that one has plenty of space left. That part makes no sense to me.”

Background: this server has three separate disks. `/srv/data` and `/srv/spool` look like ordinary folders, but each one is really its own small disk, attached at that path (a *mount point*). The rest of the system lives on a third, much larger disk.

<code>/</code>	the system disk	about 24 GB
<code>/srv/data</code>	the project data disk	1 GB
<code>/srv/spool</code>	the job-submission disk	48 MB

Each disk fills up on its own: one being full says nothing about the others. `df -h / /srv/data /srv/spool` prints one line per disk.

Though the two complaints look alike, they are **two unrelated faults on two different disks**, with different causes. Fixing one does nothing for the other.

(a) `/srv/data` is full

[11 marks]

`/srv/data` is at 100%, and the files you can see there do not add up to the space reported as used.

Reproduce:

<code>df -h /srv/data</code>	# 100% used
<code>sudo du -sh /srv/data</code>	# much smaller than df says is in use

Fix: bring `/srv/data` back to **under 10% used** (the `Use%` column of `df -h`). There is more than one reason this disk is full: freeing the largest obvious files will not finish the job. Consider the differences shown in the output of `df` and `du`. Explain the reason for the difference.

Check:

```
df -h /srv/data          # Use% under 10%
ls /srv/data/current     # five survey-block CSV files, still there
```

(b) /srv/spool refuses new files **[3 marks]**

New files cannot be created on `/srv/spool`, *although* `df -h` reports free space on it. Something other than free space has run out. This has nothing to do with `/srv/data`.

Reproduce:

```
df -h /srv/spool          # shows space free
touch /srv/spool/incoming/testjob  # "No space left on device"
```

Fix: make `/srv/spool` able to accept new files again. It is *not* short of space: freeing bytes there will not help. A filesystem can run out of more than one kind of resource, and `df` has another mode that reports the other one. Explain why `/srv/spool` reported “No space left on device” while `df -h` showed free space.

Check:

```
touch /srv/spool/incoming/t && echo OK && rm /srv/spool/incoming/t
```

Constraints

- Nothing under `/srv/data/current` may be deleted or changed. It is live survey data with no second copy on this machine.
- `/srv/data/archive/README.txt` says what is safe to remove there. Read it before deleting anything.

Q2. Shared project folder unusable. **[16 marks]**

Reported for the ACMU Lab group: “Buddhadev cannot save anything into `shared/`— he could last month. And Chandrima from the internal audit cell needs to *read* that folder for the annual review, and cannot see anything at all.”

Background: who is who. `/srv/projects/acmu/` (and `shared/` inside it) belongs to the `acmu` group. Anyone who is neither a member nor the auditor should have no access at all.

arnab	member of <code>acmu</code>	read and write everything
buddhadev	member of <code>acmu</code>	read and write everything
chandrima	auditor, not a member	read everything, write nothing

To see the current state:

```
cd /srv/projects/acmu
stat -c '%a %U:%G %n' . shared shared/notes.md
id buddhadev
getfacl /srv/projects/acmu/shared
```

Underneath the two complaints are **three independent faults**, each with its own cause and fix, each marked on its own. Take them in any order.

Test as the users named, not as root. Root can do everything, so a check run as root proves nothing.

(a) Buddhadev is no longer in the acmu group [5 marks]

Reproduce: Check `id buddhadev` where `acmu` is missing. *Fix:* put him back, without disturbing his other groups. Keep checking with `id buddhadev` until it shows `acmu` again.

(b) The shared folder does not let the group write [8 marks]

Members of `acmu` cannot create files in the folder at all. This is about the permission bits on the two directories, `/srv/projects/acmu` and `shared/`. Arnab is still a member, so test as `arnab`.

```
sudo -u arnab touch /srv/projects/acmu/shared/t1      # Permission denied
stat -c '%a %U:%G %n' /srv/projects/acmu /srv/projects/acmu/shared
```

Fix: both directories must let the `acmu` group read, write and enter them, and give everyone else no write access.

```
sudo -u arnab touch /srv/projects/acmu/shared/t1 && echo "write OK"
sudo rm -f /srv/projects/acmu/shared/t1
```

(c) The auditor cannot read the folder [3 marks]

Fix: grant read-only access to everything under `/srv/projects/acmu/`, including files created there in future, and do it *without* adding `chandrima` to the group.

```
sudo -u chandrima cat /srv/projects/acmu/shared/notes.md # must work
sudo -u chandrima touch /srv/projects/acmu/shared/t2      # must FAIL
```

When all three are done

Buddhadev's complaint needs (a) and (b) together:

```
sudo -u buddhadev touch /srv/projects/acmu/shared/t3      # must work
sudo -u buddhadev cat /srv/projects/acmu/shared/notes.md  # must work
```

Constraints

- **Chandrima must not be put into the acmu group.** Audit policy forbids putting auditors into the groups they audit: read everything, write nothing.
- No `chmod 777`, anywhere. This is a multi-user machine.

Q3. Cannot log in to the ACMU lab server. [10 marks]

Reported by Debarshi, research scholar: "I am supposed to log in to the lab server as `acmusrv`. It worked last week. Now it just says *Permission denied*, and then asks me for a password I was never given. I have not touched anything since."

`ssh acmulab` should log you in to the account `acmusrv` on the lab server using the key `~/.ssh/id_acmu`, **without a password**. Two things are wrong, and they produce the **same message**: *Permission denied (publickey)*. Identify both of them.

How to reproduce it

```
ssh acmulab

ssh -v acmulab 2>&1 | grep -i 'identity file\|offering\|publickey'
cat ~/.ssh/config
ls -l ~/.ssh/
```

What to fix

[10 marks]

3.1 Make the acmulab alias use the right key. The `~/.ssh/config` file is corrupted. Fix it so that `ssh acmulab` succeeds without a password prompt.

3.2 Make the key usable. Find out why `ssh` still refuses to use the key, and fix it.

How to check you have fixed it

```
ssh -o BatchMode=yes acmulab whoami      # prints: acmusrv
ls -l ~/.ssh/id_acmu                     # -rw----- (0600)
```

`BatchMode=yes` forbids any password prompt, so if this prints `acmusrv` the key really is doing the work.